SIMATS ENGINEERING
ASSESSMENT TOOL 3: Short Answer Test

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA1522

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COURSE OUTCOME COVERED
CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 50**
Question Count: 5

Code of Conduct
I G.Hima Bindu(192511377) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: G.Hima Bindu

Assessment Tasks
Short Answer Test
1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Compare SOAP-based and REST-based web services in terms of communication style, data exchange, and suitability for cloud applications.
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage

Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.

Definition:

Cloud Computing: The Cloud Computing is the delivery of computing services such as servers, storage ,databases,networking,software,and analytics over the over the Internet ("the cloud") on a pay-as-you-go basis. It allows users to access resources anytime without owning physical infrastructure.

Characteristics of Cloud Computing

1. On-Demand Self-Service

- Users can create or remove computing resources whenever required without contacting the service provider.
- **Example:** Launching a virtual machine in AWS within minutes.

2. Broad Network Access

- Cloud services are available over the Internet and can be accessed using laptops, smartphones, tablets, or desktops from anywhere.

3. Resource Pooling

- Cloud providers share a pool of computing resources among multiple users (multi-tenancy), improving efficiency and reducing cost.

4. Rapid Elasticity

- Resources can be increased or decreased automatically based on demand.
- **Example:** An e-commerce website increases server capacity during festival sales.

Difference from Traditional On-Premise Computing

- No need to buy and maintain hardware.
- Lower initial investment.
- Easy scalability.
- Accessible from anywhere through the Internet.

2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.

Cloud computing has developed through several stages of technological advancement.

1. Hardware Evolution(1950-1970s)

In the early days, organizations used large mainframe computers that were expensive and shared by many users through terminals. Later, powerful personal computers and servers became common.

2. Networking Evolution(1990s)

The development of computer networks allowed systems to communicate and share data. The introduction of the Internet enabled global connectivity and online communication.

3. Virtualization

Virtualization technology made it possible to run multiple virtual machines on a single physical server. This improved hardware utilization, reduced costs, and made resource management easier.

4. Internet Software Evolution

Software applications moved from desktop installations to web-based applications. Users could access software through web browsers without installing it on their computers.

5. Modern Cloud Platforms

Combining virtualization, high-speed Internet, distributed computing, and web technologies resulted in cloud platforms such as **Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP)**. These platforms provide Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS).

Conclusion

The evolution of hardware, networking, virtualization, and Internet technologies laid the foundation for modern cloud computing, making IT services more scalable, reliable, and cost-efficient.

3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.

Introduction

Server virtualization is one of the most important technologies behind cloud computing. It allows a single physical server to be divided into multiple virtual servers called **Virtual Machines (VMs)**. Each VM has its own operating system, applications, CPU, memory, and storage, even though they share the same physical hardware. This is made possible using software called a **hypervisor** (such as VMware ESXi, Microsoft Hyper-V, or KVM).

Role of Server Virtualization in Cloud Computing

1. Efficient Resource Utilization

Instead of dedicating one physical server to one application, virtualization allows multiple VMs to run on a single server. This maximizes the use of CPU, memory, and storage while reducing hardware waste.

2. Cost Reduction

Organizations save money because fewer physical servers are needed. This reduces:

- Hardware purchase costs
- Electricity consumption
- Cooling costs
- Maintenance expenses

3. Scalability

Virtual machines can be created, resized, or deleted quickly based on user demand. This enables cloud providers to easily scale services up or down.

4. Isolation and Security

Each virtual machine operates independently. If one VM crashes or is infected by malware, other VMs remain unaffected.

5. Faster Deployment

Creating a new virtual server takes only a few minutes, whereas setting up a physical server may take several hours or days.

6. Disaster Recovery and Backup

Virtual machines can be easily backed up, cloned, and restored, making disaster recovery faster and more reliable.

7. High Availability

If one physical server fails, VMs can be moved to another server with minimal downtime, ensuring continuous service.

Virtualization as an Enabling Technology vs Cloud Computing as a Service Model

Virtualization	Cloud Computing
A technology that creates virtual machines.	A service model that delivers computing resources over the Internet.
Uses a hypervisor to divide physical servers into multiple VMs.	Uses virtualization and other technologies to provide cloud services.
Focuses on efficient use of hardware.	Focuses on delivering services to users.
Can be implemented inside a company's own data center.	Services are usually provided by cloud providers over the Internet.
Users manage the virtual machines themselves.	Cloud providers manage the infrastructure.
Example: VMware ESXi, Hyper-V, KVM.	Example: AWS, Microsoft Azure, Google Cloud Platform.

Relationship Between Virtualization and Cloud Computing

- Virtualization is the **foundation** of cloud computing.
- Cloud computing uses virtualization to create and manage virtual servers efficiently.
- Without virtualization, cloud providers would need separate physical servers for every customer, making cloud services expensive and difficult to manage.

Real-World Example

Amazon Web Services (AWS) uses virtualization technology to create thousands of virtual machines (EC2 instances) on powerful physical servers. Customers can launch or stop these virtual machines whenever they need them and pay only for the resources they use.

Conclusion

Server virtualization is the backbone of cloud computing because it improves resource utilization, reduces costs, increases flexibility, and enables cloud providers to deliver scalable, reliable, and on-demand services. Virtualization is the enabling technology, while cloud computing is the service model built on top of it.

4. Compare SOAP-based and REST-based web services in terms of communication style, data exchange, and suitability for cloud applications.

Introduction

Web services allow different applications to communicate over a network. The two most widely used web service technologies are **SOAP (Simple Object Access Protocol)** and **REST (Representational State Transfer)**.

SOAP (Simple Object Access Protocol)

SOAP is a protocol that uses XML for exchanging structured information between applications. It follows strict standards and supports advanced security features.

Features

- Uses only **XML** for data exchange.
- Platform-independent.
- Supports WS-Security, digital signatures, and encryption.
- Reliable messaging with built-in error handling.
- Can work over HTTP, SMTP, TCP, and other protocols.

Advantages

- High security.
- Reliable message delivery.
- Suitable for financial transactions and enterprise systems.
- Supports ACID transactions.

Disadvantages

- Complex to implement.
- XML messages are large, increasing bandwidth usage.
- Slower than REST.

REST (Representational State Transfer)

REST is an architectural style that uses standard HTTP methods for communication.

Features

- Uses HTTP methods such as GET, POST, PUT, DELETE.

- Supports JSON, XML, HTML, and plain text.
- Stateless communication.
- Lightweight and easy to develop.

Advantages

- Fast performance.
- Lower bandwidth usage.
- Easy integration with mobile and web applications.
- Highly scalable.

Disadvantages

- No built-in security standard like SOAP.
- Security depends on HTTPS, OAuth, JWT, etc.

Comparison of SOAP and REST

Feature	SOAP	REST
Full Form	Simple Object Access Protocol	Representational State Transfer
Type	Protocol	Architectural Style
Data Format	XML only	JSON, XML, HTML, Text
Communication	Protocol-based	HTTP-based
Speed	Slower	Faster
Message Size	Large	Small
Security	WS-Security	HTTPS, OAuth, JWT
Flexibility	Less flexible	Highly flexible
Performance	Lower	Higher
Best Used For	Banking, Healthcare, Enterprise	Cloud APIs, Mobile Apps, Web Services

Suitability for Cloud Applications

SOAP is suitable for:

- Banking systems
- Online payment gateways
- Healthcare applications
- Enterprise business applications
- Secure government services

REST is suitable for:

- Cloud computing platforms
- Mobile applications
- Social media applications
- E-commerce websites
- IoT devices
- Public APIs

Real-World Examples

- **SOAP:** Banking transaction systems and PayPal APIs (legacy services).
- **REST:** AWS REST APIs, Google Maps API, Twitter API, GitHub API.

Conclusion

SOAP is best when strong security, reliability, and transaction support are required. REST is preferred for cloud computing because it is lightweight, faster, scalable, and easier to develop. Most modern cloud applications use REST APIs.

5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Introduction

Cloud computing provides different service models based on how much responsibility the cloud provider and the customer have. The four major service models are **IaaS, PaaS, SaaS, and XaaS**.

1. IaaS (Infrastructure as a Service)

Definition

IaaS provides virtualized computing resources such as servers, storage, networking, and virtual machines over the Internet.

Features

- Virtual servers
- Storage
- Networking
- Load balancing
- High scalability
- Pay-as-you-use pricing

Advantages

- No need to buy physical servers.
- Easy to scale.
- Full control over the operating system and applications.

Example

Amazon EC2 (AWS) – Users rent virtual machines and install their own operating systems and software.

2. PaaS (Platform as a Service)

Definition

PaaS provides a complete platform for developing, testing, deploying, and managing applications without managing the underlying infrastructure.

Features

- Development tools
- Runtime environment
- Databases
- Middleware
- Automatic software updates

Advantages

- Faster application development.
- No server management.
- Supports team collaboration.

Example

Google App Engine – Developers upload code, and Google manages the servers and operating system.

3. SaaS (Software as a Service)

Definition

SaaS delivers ready-to-use software applications through the Internet. Users simply log in and use the software without installing or maintaining it.

Features

- Browser-based access
- Automatic updates
- Subscription model
- Multi-user access

Advantages

- No installation required.
- Accessible from anywhere.
- Low maintenance.

Example

Google Workspace (Gmail, Docs, Sheets) – Users access office applications through a web browser.

4. XaaS (Anything as a Service)

Definition

XaaS stands for **Anything (or Everything) as a Service**. It is a broad term that includes all cloud-based services delivered over the Internet.

Examples of XaaS

- Storage as a Service (STaaS)
- Database as a Service (DBaaS)
- Network as a Service (NaaS)
- Security as a Service (SECaaS)
- Backup as a Service (BaaS)

Advantages

- Flexible
- Cost-effective
- Easy to integrate
- Highly scalable

Example

Microsoft 365 or **Dropbox** provides cloud-based productivity and storage services.

Comparison Table

Feature	IaaS	PaaS	SaaS	XaaS
Full Form	Infrastructure as a Service	Platform as a Service	Software as a Service	Anything as a Service
Provides	Infrastructure	Development Platform	Ready-to-use Software	Any cloud service
User Manages	OS, Apps, Data	Apps & Data	Only uses the software	Depends on the service
Provider Manages	Hardware, Network	Infrastructure + OS	Everything	Varies
Users	System Administrators	Developers	End Users	Individuals & Organizations
Example	Amazon EC2	Google App Engine	Google Workspace	Microsoft 365 / Dropbox

Real-World Scenario

A company wants to build and use an online shopping website:

- **IaaS:** Rent virtual servers from AWS EC2.
- **PaaS:** Develop the website using Google App Engine.

- **SaaS:** Employees use Google Workspace for email and document sharing.
- **XaaS:** Store backups using Dropbox or use Database-as-a-Service for customer records.

Conclusion

IaaS, PaaS, SaaS, and XaaS provide different levels of cloud services. IaaS offers infrastructure, PaaS provides a development platform, SaaS delivers ready-to-use software, and XaaS represents any service delivered through the cloud. Organizations choose the model based on their technical needs, budget, and level of control required.